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PEER: Unified Process–Outcome Reinforcement Learning for Structured Empathetic Reasoning

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Abstract

Emotional support conversations require more than fluent responses. Supporters need to understand the seeker's situation and emotions, adopt an appropriate strategy, and respond in a natural, human-like manner. Despite advances in large language models, current systems often lack structured, psychology-informed reasoning. Additionally, it is challenging to enhance these systems through reinforcement learning because of unreliable reward signals. Moreover, reinforcement fine-tuning can amplify repetitive response patterns. We propose ***structured empathetic reasoning***, which breaks support into three steps: conversation history analysis, multimodal emotional state inference, and strategy selection, prior to generating the final reply. To implement this, we introduce SER, a fine-grained dataset with step-level correctness labels and pairwise response preferences. We then present **PEER**, which uses GRPO with *UnifiReward*, a unified process–outcome reward model for evaluating both reasoning steps and final responses in multi-turn interactions. To reduce repetition, we enhance data with personality-based rewriting and down-weight redundant outputs. Comprehensive experiments show improved empathy, strategy alignment, and human-likeness without sacrificing diversity.

Keywords

Emotional Support Conversations, Psychology-Informed Reasoning

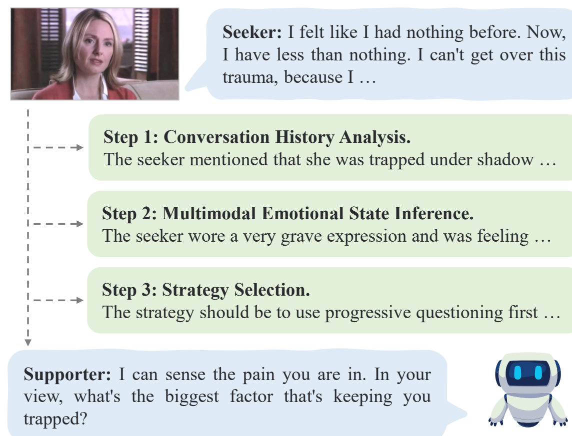


Figure 1: Structured empathetic reasoning.

1 Introduction

Emotional Support Conversation (ESC) aims to comfort people experiencing emotional distress through natural dialogue [16]. With Large Language Models (LLMs), ESC systems have improved substantially in fluency and general helpfulness [21]. Recently, visual information, including facial expressions, has also been incorporated to achieve multimodal ESC [6]. However, producing *effective* emotional support still requires more than writing a polite response. A supporter must infer the seeker's situation and emotional state, choose an appropriate support strategy, and respond in a human-like manner. In practice, even strong LLMs frequently default to generic empathy, premature advice, or strategy-inconsistent responses, which limits their usefulness in real support settings.

A central reason is that ESC relies on higher-order social cognition rather than formal logical reasoning. Recent reasoning-oriented training has improved LLMs performance on structured tasks [17], but these gains do not reliably transfer to psychologically grounded dialogue behaviors [30]. Existing ESC methods [6, 36] partially address this gap by adding emotion/strategy annotations or prompting models to produce rationales. However, two limitations still persist. First, standard supervised fine-tuning does not directly supervise *why* a response is appropriate. A response can appear plausible even if the model misreads the seeker's emotion or chooses an unsuitable strategy. Second, improving ESC models with Reinforcement Learning (RL) is challenging. Emotional support is open-ended, making rule-based rewards brittle, while off-the-shelf LLM judges are often unreliable at evaluating psychologically grounded intermediate reasoning. Moreover, RL fine-tuning can amplify repetitive patterns (entropy collapse) [7, 38], reducing diversity and making conversations feel mechanical.

As illustrated in Figure 1, we address these challenges with ***structured empathetic reasoning***, a lightweight reasoning scaffold that decomposes emotional support into three steps before generating final response: 1) *Conversation history analysis* to identify the seeker's issue and the current support stage, 2) *Multimodal emotional state inference* to assess affect based on both facial expressions and textual utterances, and 3) *Strategy selection* to determine the appropriate response. This structure makes the model's supportive behavior more interpretable and provides natural supervision targets beyond final utterance.

To support this capability, we first construct the **Structured Empathetic Reasoning (SER)** dataset, which is fine-grained annotated with 1) step-level correctness labels for the three reasoning steps and 2) pairwise response preferences for the final reply. These annotations provide learning signals for both the reasoning process and response quality. Because diversity is crucial in ESC, we further augment SER with personality-based conversation rewriting, which increases stylistic coverage while preserving the original supportive intent.

Building on SER, we then propose **emPathetic rEinforcEment Reasoning (PEER)** framework that improves both reasoning fidelity and response quality. PEER uses GRPO [23] as the underlying optimizer and introduces *UnifiReward*, a **unified process–outcome reward model** that jointly evaluates intermediate reasoning steps (process) and the final response (outcome) in multi-turn settings. Unifying these signals within a single reward model diminishes the inconsistencies that occur when separate process and outcome evaluators reach different conclusions. Finally, to mitigate RL-induced repetition, we incorporate a **redundancy-aware reward reweighting** mechanism that down-weights overly similar candidate outputs relative to the conversation history and peer generations. Experiments on ESC benchmarks and human preference evaluation show that PEER improves empathy, strategy alignment, and human-likeness while maintaining response diversity.

Our contributions can be summarized as follows:

- We propose structured empathetic reasoning, a psychologically grounded reasoning scaffold for ESC, and build SER, a dataset annotated with step-level correctness and response preferences to support this capability.
- We introduce PEER, an RL framework that uses UnifiReward to deliver consistent, fine-grained feedback in multi-turn dialogues.
- We mitigate RL-induced repetition via personality-based rewriting and redundancy-aware reward reweighting, improving diversity without sacrificing empathy or coherence.

2 Related Work

2.1 Emotional Support Conversation

ESC plays a crucial role in advancing the application of LLMs in fields such as psychological counseling and social forums. Liu et al. [16] introduced the first ESC dataset, ESConv, by creating manually crafted conversations, which significantly promoted research on ESC systems. Meanwhile, Chu et al. [6] incorporated visual information, such as facial expressions, to support multimodal ESC. However, the manual construction of ESC datasets is costly and labor-intensive, resulting in limited dataset sizes. To address this limitation, recent studies have focused on leveraging LLMs to expand conversations based on real single-turn psychological counseling reports [13, 36] or character-based prompts [4, 25, 27], thereby generating larger datasets. Nevertheless, the dialogues generated by LLMs often lack stylistic flexibility and deep reasoning. ESCoT [33] incorporates a chain-of-thought approach to enhance the model's emotional support capabilities. However, it lacks annotations for reasoning correctness and outcome preference, which limits its applicability in reinforcement learning frameworks. We address this issue by refining the structure of the thought chain and providing manual annotations for both the reasoning process and the final outcomes. Furthermore, the majority of existing works [3, 6, 27, 31] solely utilize supervised fine-tuning during model training. This training approach is overly simplistic to comprehensively tap into the potential of the existing models and datasets. As a result, we incorporate reinforcement learning algorithms to train the model and optimize it for emotional support scenarios.

2.2 RL with Reward Models

RL enables models to learn through trial-and-error interaction with an environment. Unlike supervised learning, RL naturally handles non-differentiable feedback and optimizes complex objectives [15]. In LLMs, RL was first widely used for human preference alignment, most notably via Reinforcement Learning from Human Feedback (RLHF) [20].To simplify RLHF, Direct Preference Optimization (DPO) [22] bypasses reward modeling and directly optimizes the policy from preference pairs, improving efficiency and stability. More recently, RL has been applied to enhance LLM reasoning [17]. For example, GRPO [23] is designed to strengthen reasoning performance through group-based relative advantage estimation, which helps alleviate the effects of noisy or sparse reward signals and is particularly beneficial for challenging reasoning tasks such as mathematical problem solving. However, recent studies [2, 7, 38] shows RL training can cause entropy collapse, pushing outputs toward overly rigid pattern. These findings highlight both the promise of RL and the importance of carefully designing training objectives and regularization strategies.

Reward models play a central role in shifting LLM training from passive learning on static datasets to active optimization under dynamic feedback [28, 29]. From the perspective of reward granularity, existing reward models can be broadly categorized into Outcome Reward Models (ORMs), which assess the quality of final answers, and Process Reward Models (PRMs), which evaluate intermediate reasoning steps in a fine-grained manner. These two paradigms provide complementary supervision signals: ORMs emphasize task-level correctness, whereas PRMs offer richer guidance over the reasoning trajectory. Their integration therefore has the potential to provide a more comprehensive reward signal and to further improve reasoning quality. Nevertheless, combining them also introduces new challenges, particularly when process-level assessments and final-outcome evaluations are not fully consistent.

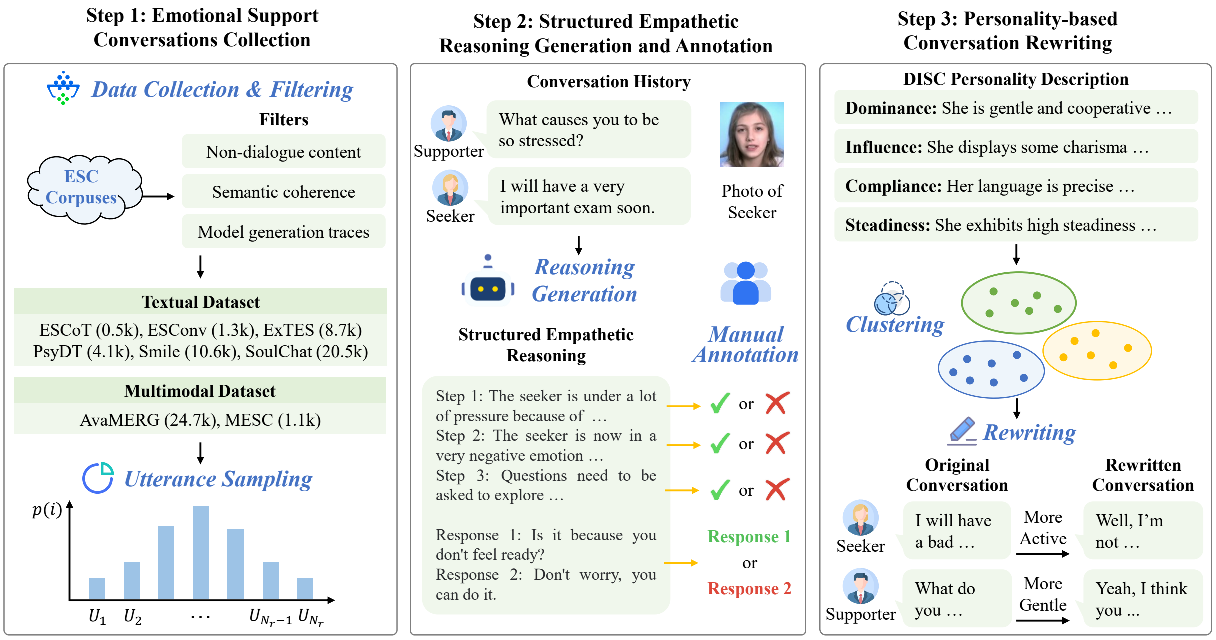


Figure 2: Construction pipeline of the SER dataset.

3 Our SER Dataset

3.1 Conversation Collection

As shown in Figure 2, we first compile a comprehensive collection of open-source emotional support conversation datasets, including multimodal datasets (MESC [6], AvaMERG [32]) and textual corpora (ESConv [16], SoulChat [3], PsyDTCorpus [27], CPsyCoun [31], SmileChat [21], ExTES [37], ESCoT [33]). These datasets cover both Chinese and English and span a wide range of emotional support scenarios, thereby providing a strong foundation for subsequent training and analysis. However, some of the collected data contain dialogues generated by LLMs, which often exhibit recognizable machine-generated patterns. To improve data quality, we employ Qwen2.5-72B as a filtering model, using the prompt shown in Figure 2 of the supplementary material to identify and remove samples with semantic incoherence, artificial stylistic patterns, or non-dialogue content. This filtering step removes more than 23% of the data, substantially improving the reliability of the retained corpus. A qualitative example of a filtered dialogue is presented in Figure 3 of the supplementary material, where a large amount of explanatory text is enclosed in parentheses and clear signs of AI-generated content are observed.

To reduce redundancy caused by strong contextual overlap between adjacent utterances, we adopt sparse sampling to retain the most informative content. For a dialogue with N_r rounds, we select N_s sentences according to the following Gaussian distribution:

$$p(i) = \frac{\exp\left(-\frac{(i-\mu)^2}{2\sigma^2}\right)}{\sum_{j=1}^{N_r} \exp\left(-\frac{(j-\mu)^2}{2\sigma^2}\right)}, \quad (1)$$

where $p(i)$ denotes the probability of sampling the i-th round, with $\mu = N_r/2$ (favoring the middle of the dialogue) and $\sigma = N_r/4$ (ensuring moderate spread). This design reflects the observation that central dialogue rounds are more content-rich and emotionally meaningful, while the start and end often contain generic pleasantries (e.g., “hello”, “goodbye”). We then sample N_s unique indices from this distribution to obtain the final subset $\mathcal{U} = \{u_i\}_{i=1}^{N_s}$, while preserving their original order.

3.2 Structured Empathetic Reasoning Generation and Annotation

To address the limitations of existing models, we propose a structured empathetic reasoning framework, as illustrated in Figure 1. This framework integrates established psychological principles from prior ESC systems [33] with human-like reasoning mechanisms commonly adopted in recent reasoning-oriented models [34]. Specifically, we decompose the reasoning process into three structured stages: *1) Dialogue history analysis.* This stage identifies the seeker's concerns from the dialogue context and determines the current stage of emotional support, following the three-phase structure of exploration, comforting, and action defined in ESConv [16]. *2) Multimodal emotional state inference.* This stage infers the seeker's emotional state from both facial expressions and textual utterances, including coarse-grained sentiment (positive, negative, and neutral), fine-grained emotions (e.g., joy and fear), and sentiment intensity. *3) Strategy selection.* Based on Hill's Helping Skills Theory [10], this stage selects the most appropriate support strategy from eight predefined categories, such as progressive questioning and emotional mapping. Overall, the framework enables the model to follow psychologically grounded reasoning steps while preserving the flexibility of natural language generation.

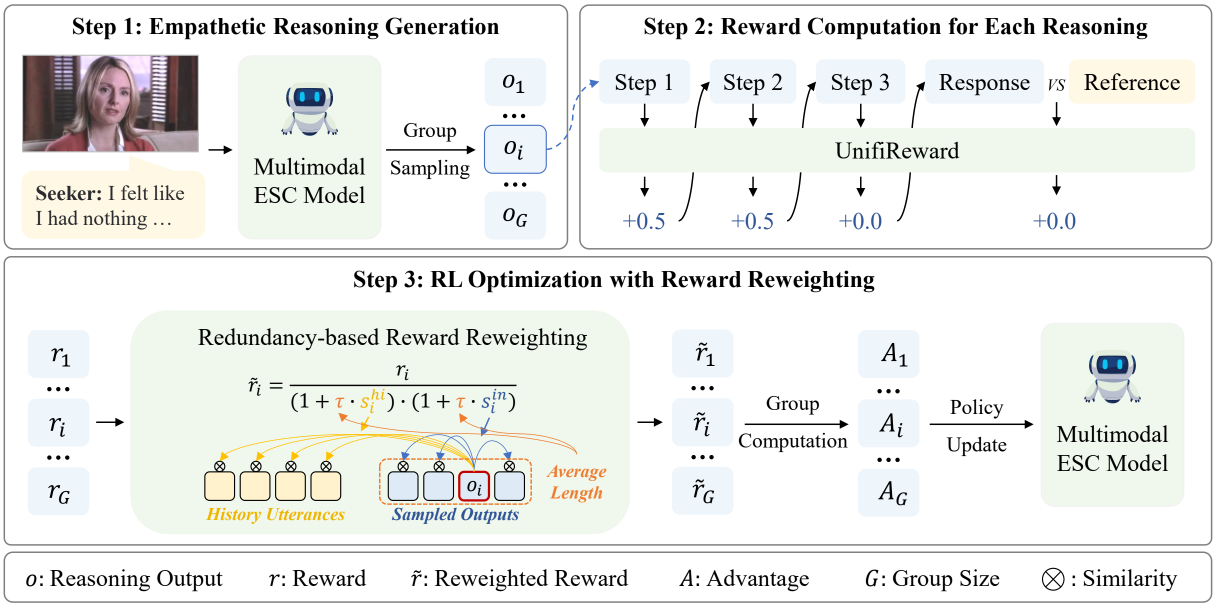


Figure 3: Illustration of our PEER framework.

Existing ESC datasets typically lack both negative samples and intermediate reasoning annotations that explicitly support structured empathetic reasoning. To address this gap, we construct and **manually** annotate the SER dataset. Specifically, we input the conversation history composed of sampled sentences into GPT-4o [19] and generate candidate responses using the prompts in Figures 4 and 5 of the supplementary material. These prompts include the following key components: 1) an explanation of the three stages of emotional support; 2) a definition of candidate emotional states; 3) a description of candidate affective strategies; and 4) the required output format.

To ensure annotation quality, we **recruited 20 trained annotators through a crowd sourcing platform** and provided reasonable compensation to all participants. Detailed annotation guidelines are presented in Figure 6 of the supplementary material. Each annotated sample consists of two auxiliary fields (History and Photos) and two target annotation fields (Steps and Responses). Because the two annotation tasks differ in nature, with reasoning correctness being relatively objective and response quality being more subjective, we adopt different annotation protocols for each. For empathetic reasoning steps, we apply point-wise annotation and assign a binary label ("True" or "False") to each step. For responses, we use pair-wise preference labeling by comparing the original response with the model-generated alternative. Annotators choose from four options: "The first sentence is better", "The second sentence is better", "The two sentences are almost good", and "Neither sentence is acceptable".

Each example was initially labeled independently by two annotators. In cases of disagreement, a third annotator was introduced, and the final label was determined by majority voting. Samples that failed to reach consensus, defined as agreement between at least two annotators, were excluded from the final dataset. This process yielded 22,240 high-quality annotations across 5,560 dialogues. Among them, 4,986 dialogues are used for training and 574 for testing. The Fleiss kappa score is 0.69 for the retained step labels and 0.56 for the retained response labels, indicating moderate to substantial agreement. The label distributions for both steps and responses are shown in Figure 1 of the supplementary material, and two annotation examples are provided in Figure 7 and Figure 8 of the supplementary material.

3.3 Personality-based Conversation Rewriting

Text rewriting is an effective data augmentation strategy for LLM training [9, 24]. However, in our setting, naive dialogue rewriting fails to improve model accuracy. Further analysis suggests that rewritten dialogues often converge toward a homogeneous linguistic style, likely reflecting the inherent personality tendencies of the rewriting LLM, a phenomenon consistent with prior findings [11]. In addition, previous work [25] has shown that personality traits can strongly influence the selection of emotional support strategies. Consequently, personality drift introduced during rewriting may become misaligned with the original strategy annotations, thereby weakening the effectiveness of the training signal.

To address this, we propose a personality-based conversation rewriting approach. We first extract personality traits of dialogue participants using DISC Behavior Pattern Theory [18], which categorizes communication behaviors into four types: Dominance (D), Influence (I), Steadiness (S), and Compliance (C). GPT-4o is used for trait extraction under the guidance of the structured prompt shown in Figure 9 of the supplementary material. We then apply K-Means clustering to group the extracted personality descriptions and determine the optimal number of clusters using the elbow method based on the sum of squared errors. For each dialogue, we randomly sample a personality profile from the same cluster and use GPT-4o to rewrite the dialogue so that it aligns with the new persona, following the prompt in Figure 10 of the supplementary material. Finally, we apply an additional filtering step using the prompt in Figure 11 of the supplementary material to remove rewritten samples that exhibit semantic inconsistencies or detectable LLM artifacts, thereby ensuring consistency with the original dialogue intent.

4 Our Proposed PEER

Unlike supervised fine-tuning, RL enables models to actively improve through dynamic feedback by leveraging both positive and negative signals, thereby reducing the likelihood of repeatedly making the same errors. Motivated by this advantage, we employ GRPO [23] to fine-tune our PEER model, as shown in Figure 21. In this framework, PEER serves as the policy model and is optimized with the aim of maximizing the reward provided by UnifiReward. To alleviate entropy collapse, the obtained rewards are reweighted based on redundancy to encourage diverse outputs.

4.1 UnifiReward

To provide fine-grained supervision, we introduce UnifiReward, a unified reward model that supports multi-turn evaluation and enables both step-wise feedback, similar to PRM, and outcome-level comparisons, similar to ORM. Inspired by recent generative reward modeling frameworks [8, 14], UnifiReward outputs natural language judgments, which are subsequently parsed into actionable scalar rewards. Specifically, for the t-th round of the i-th sample, the reward model outputs a token sequence $\mathbf{y}_t^i = [v_1^{(t,i)}, v_2^{(t,i)}, ..., v_{n_t^i}^{(t,i)}]$, where n_t^i represents the total number of tokens. The reward model is trained by minimizing the following cross-entropy loss:

$$\mathcal{L}_t^i(\theta_r) = -\sum_{j=1}^{n_t^i} \log P\left(v_j^{(t,i)} \mid H_t^i \quad \oplus [v_1^{(t,i)}, \dots, v_{j-1}^{(t,i)}]; \theta_r \ \right), \tag{2}$$

where θ_r is the parameters of the reward model, H_t^i represents the conversation history, and $\oplus$ denotes token concatenation.

UnifiReward assigns rewards according to three criteria: 1) Reasoning step accuracy: +0.5 for each step that gets “True” from UnifiReward and 0 for each step that gets “False”. 2) Final response preference: +0.5 if the output matches or outperforms the reference; 0 otherwise. For example, if the output of PEER is in the first position, +0.5 when UnifiReward outputs “The first sentence is better” or “The two sentences are almost equally good”, and 0 in all other cases. 3) Format completeness: +1 if all steps and final response are included; 0 if any are missing.

Under this design, reasoning correctness can contribute up to 2 points in total, whereas format completeness contributes up to 1 point. In this way, reasoning quality is weighted twice as heavily as format compliance, encouraging the model to prioritize accurate and well-structured reasoning rather than merely producing outputs in the required format. Overall, this unified scoring mechanism provides precise and interpretable feedback, thereby supporting the joint optimization of reasoning quality and response effectiveness during RL training. More detailed scoring rules are provided in Figure 12 of the supplementary material.

4.2 Redundancy-based Reward Reweighting

Although reinforcement learning improves the quality of emotional support generation, we observe that the model tends to produce increasingly repetitive responses in the later stages of training, which reduces output diversity. This phenomenon is consistent with the entropy collapse behavior reported in prior studies [2, 38]. To alleviate this issue, we propose a redundancy-aware reward reweighting strategy that down weights the rewards of outputs exhibiting high similarity either to other generated responses or to the conversation history.

For an output o_i, its reward r_i is adjusted as follows:

$$\tilde{r}_i = \frac{r_i}{(1 + \tau \cdot s_i^{hi}) \cdot (1 + \tau \cdot s_i^{in})}, \tag{3}$$

where $\tilde{r}_i$ is the reweighted reward, s_i^{in} measures the in-group redundancy, namely the similarity between o_i and other generated outputs in the same group, and s_i^{hi} captures history redundancy, namely the similarity between o_i and the conversation history. These two terms are computed as:

$$\begin{cases} s_i^{\text{in}} &= \frac{1}{G-1} \sum_{j \neq i} \text{Cos}(o_i, o_j), \\ s_i^{\text{hi}} &= \frac{1}{|H|} \sum_{u \in H} \text{Cos}(o_i, u), \end{cases} \tag{4}$$

where u denotes an utterance in the conversation history, $|H|$ represents the total number of utterances in the history, and $\text{Cos}(\cdot, \cdot)$ denotes the cosine similarity.

To avoid over-penalizing short responses, especially those that are structurally natural at dialogue boundaries (e.g., “hello”, “goodbye”), we further introduce a length-based coefficient τ to modulate the redundancy penalty:

$$\tau = \frac{1}{1 + e^{-\alpha(L-\beta)}}, \tag{5}$$

where L is the average length of all output sequences, α (> 0) controls the steepness of the transition, and β sets the threshold. When $L \ll \beta$, τ approaches 0, thereby minimizing the penalty for brief responses. In contrast, when $L \gg \beta$, τ approaches 1, allowing the full penalty to take effect. In this way, the proposed adaptive mechanism encourages diversity while avoiding unnecessary penalization of brief yet contextually appropriate utterances.

5 Experiments

5.1 On Reward Model

5.1.1 Experimental Settings. We adopted Qwen3-VL-8B as the backbone of our reward model and fine-tuned it on the SER dataset. Specifically, the LLM and aligner parts are optimized using LoRA with a rank of 16 and an alpha value of 64. The learning rate is set to 2×10^{-4}, and the batch size is 32. All experiments are conducted on 8 NVIDIA A800 GPUs.

To evaluate the reward model, we used test-set of the SER dataset consisting of 574 samples. For the three intermediate reasoning steps, which are formulated as binary classification with class imbalance (23% “False”), we reported Weighted-F1 and Recall

Table 1: Performance comparison and ablation study of the reward models on the SER dataset (mean±std). Best results in bold; second-best underlined. All bold results significantly outperform prior baselines ($p < 0.05$).

Model	Step 1		Step 2		Step 3		Response	
	Weighted-F1	Recall	Weighted-F1	Recall	Weighted-F1	Recall	Weighted-F1	Accuracy
GPT-4o	46.63±0.03	6.48±0.28	75.53±0.78	4.26±0.25	73.42±0.74	10.91±0.89	47.94±0.31	51.92±0.92
InternVL3.5	45.84±0.47	2.94±0.42	76.65±0.11	8.00±0.50	71.77±0.27	2.68±0.89	45.97±0.08	52.26±0.04
MiMo-VL	47.78±0.05	7.20±2.94	75.11±1.05	9.06±0.50	71.64±0.40	6.14±3.56	48.15±1.01	53.66±0.35
Keye-VL1.5	54.26±0.44	15.55±0.42	73.58±1.24	17.50±1.00	72.74±0.69	16.97±2.68	42.42±2.28	47.04±3.48
Qwen3-VL	54.83±0.34	19.33±1.68	72.06±0.24	19.00±0.35	72.13±1.31	23.22±3.58	44.49±0.97	50.52±1.04
PRM	71.39±2.11	54.68±2.10	75.61±0.02	20.25±1.25	74.83±0.31	21.43±2.68	-	-
ORM	-	-	-	-	-	-	45.21±1.48	51.08±2.27
UnifiReward	71.40±0.66	55.46±3.36	78.58±1.13	21.50±1.50	75.83±0.49	25.15±2.14	49.76±0.38	53.84±0.18
w/o Visual	70.49±1.52	54.21±2.31	76.24±2.05	18.85±1.25	74.22±1.31	20.67±2.68	48.55±1.48	49.66±1.05
Rw w/o Person	60.42±2.91	30.25±4.04	77.25±0.46	13.00±3.00	75.21±2.61	21.43±2.93	49.46±1.01	51.05±0.53
Rw w. Person	**77.58**±0.76	**67.23**±2.04	**81.18**±0.75	**30.00**±1.75	**79.15**±2.03	**42.86**±1.93	**60.12**±1.38	**62.15**±0.87

of the "False" class. For final response evaluation, which is formulated as a four-class classification task, we reported Weighted-F1 and Accuracy. Significance testing was via paired bootstrap test (10,000 resamplings).

We compared UnifiReward with several MLLMs, including GPT-4o [19], InternVL3.5 [5], Qwen3-VL [1], Keye-VL1.5 [12] and MiMo-VL [26]. Except for GPT-4o, all baseline models fall within the 7B–8B parameter range and are evaluated using the same prompt for a fair comparison. To evaluate the benefits of unified reward modeling, we further trained two specialized variants, namely a PRM and an ORM, using the same training data as UnifiReward. In addition, to assess the effect of our proposed personality-based conversation rewriting strategy, we considered two data augmentation variants: **w/o Visual**, which eliminates facial imagery; **Rw w/o Person**, which applies dialogue rewriting without personality constraints; and **Rw w. Person**, which applies personality-aware rewriting, namely our full method. All variants are trained with the same number of samples.

5.1.2 *Performance Comparison.* As shown in Table 1, our reward model outperforms general-purpose models across all evaluation components. This result underscores the challenge faced by general MLLMs in reliably evaluating high-level social-cognitive tasks, particularly those involving psychological reasoning, and support strategy selection. In contrast, fine-tuning on our SER dataset substantially improves performance, especially in subtle judgment scenarios that require psychologically grounded reasoning.

Among the reward-model variants, UnifiReward achieves better performance than both PRM and ORM across all evaluation steps. This advantage likely stems from its unified multi-turn design, which allows the model to leverage information from preceding reasoning stages to improve judgment quality in subsequent steps. After eliminating the facial images, the recall rate of each step decreased significantly, which indicates that multimodal information is helpful in identifying reasoning errors. In addition, compared with personality-based rewriting, the variant trained with vanilla rewritten dialogues without explicit personality constraints exhibits a clear performance drop. A plausible explanation is that LLM-based rewriting tends to reflect the model's inherent stylistic preferences, which may alter the distribution of affective strategies and reduce consistency with the intended psychological framework.

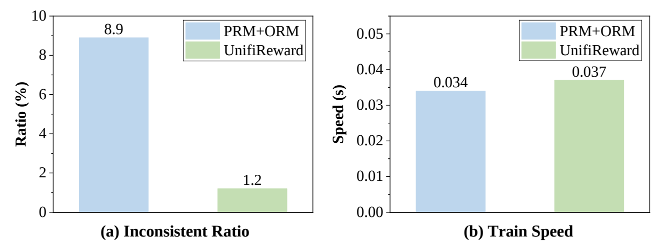


Figure 4: Comparison of inconsistency ratios and training speed across different reward model formats.

5.1.3 *Consistency and Efficiency Study.* We observed that reward models may produce inconsistent process-level and outcome-level evaluations. For example, a reasoning chain may correctly identify the need for emotional soothing, while the selected response may instead focus on offering practical suggestions. To quantify this issue, we compare UnifiReward with the PRM+ORM combination. Specifically, we collect 1,000 samples through rejection sampling and use GPT-4o to assess the consistency between the reasoning process and the final response. As shown in Figure 4 (a), UnifiReward exhibits fewer inconsistent cases. This is likely because PRM and ORM make judgments independently, while UnifiReward evaluates the outcome in light of the reasoning process, promoting coherence across stages.

During GRPO training, the reward model evaluates each output online. Although the multi-turn structure of UnifiReward introduces additional evaluation steps, the resulting computational overhead remains small. As shown in Figure 4 (b), UnifiReward increases the average training time per round by only 0.003 seconds, corresponding to an 8.8% overhead compared with PRM+ORM.

Table 2: Rewritten dialogue quality assessment (mean±std).

Model	SC	CR	RC	LF
GPT-4o	2.95±0.02	2.99±0.03	2.81±0.01	2.99±0.01
DeepSeek-V3	2.99±0.01	2.98±0.02	2.88±0.04	2.99±0.01

Considering the gains in both consistency and reward accuracy, this additional cost is modest and practically acceptable.

5.1.4 Quality of Rewritten Dialogues. To rigorously assess the quality of dialogues generated by our personality-driven rewriting approach, we used GPT-4o and DeepSeek-V3 as evaluator models for a multi-dimensional analysis. The evaluation focuses on four key aspects: Semantic Consistency (SC), Contextual Rationality (CR), Role Consistency (RC) and Language Fluency (LF). Each dimension is scored on a scale from 0 to 3, the scoring criteria are presented in Table 1 of the supplementary material, and the evaluation results are reported in Table 2.

As shown in the table, the two evaluator models produce highly consistent scores across all dimensions, supporting the reliability of the evaluation results. Moreover, the rewritten dialogues achieve near-ceiling performance on all four dimensions, indicating that the proposed rewriting method produces high-quality dialogues while preserving both semantic fidelity and stylistic appropriateness. This strong performance can be attributed, at least in part, to the effectiveness of our filtering mechanism, which removes low-quality or inconsistent rewritten samples before training.

5.2 On Emotional Support

5.2.1 Implementation Details. We also adopted Qwen3-VL-8B as the backbone model of our emotional support framework. During training, the ViT module is frozen, while the LLM and aligner modules are fully fine-tuned. The learning rate is set to 1×10^{-6}, the batch size is 24, and the group size in GRPO is 4. In the redundancy-based reward reweighting module, α is set to 0.5 and β is set to 5. In addition, we used jina-embeddings-v3 to extract response embeddings for redundancy estimation. Significance testing was via paired bootstrap test (10,000 resamplings).

5.2.2 Evaluation Benchmarks. We adopted the following automated evaluation frameworks to assess the emotional support capabilities:

1) ESC-Eval [35] contains 656 scenarios, each prompting a five-round dialogue between ESC-Role and the evaluated model. The resulting dialogues are scored by both GPT-4o and DeepSeek-V3 across six dimensions (scoring criteria in Supplementary Tables 2 and 3), and final scores are averaged to mitigate evaluator bias.

2) SAGE [30] simulates up to 20 rounds of dialogue between a seeker agent and the ESC model. After each round, it computes the seeker's sentient score to dynamically assess emotional support quality. The dialogue ends early if the score exceeds 100 (success) or drops below 10 (failure). Performance is reported as the mean of final sentient scores across 100 test scenarios, using both GPT-4o and DeepSeek-V3 as seeker agents to minimize assessment bias.

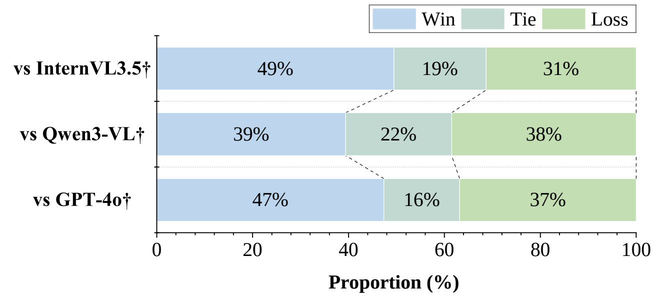


Figure 5: Human evaluation results comparing our model with baseline models on multimodal dialogues. †: utilizing structured three-stage prompt.

5.2.3 Baselines. We evaluated two categories of baseline models: 1) General-purpose models, including InternLM3, Qwen3, InternVL3.5, and Qwen3-VL; 2) Emotion-specialized models, including CPsyCounX, EmoLLMV3, and SoulChat2. To enable a comprehensive comparison, we reported two sets of results for general-purpose models: one using a vanilla prompt, and the other using the structured three-stage prompt derived from our framework. The vanilla prompt is: "You are an empathetic chat companion who communicates intelligently, making users feel at ease and supported. Respond in a natural, conversational tone—keep replies simple, human, and free of technical jargon". Due to compatibility constraints, prompt-aligned results are not reported for certain proprietary models, such as SoulChat2, which cannot be adapted to our prompting scheme. All baseline models are within the 7B–8B parameter range. InternLM3 and Qwen3 are evaluated in deep-reasoning mode, whereas the other models generate direct responses.

To assess the contributions of each component in our framework, we further constructed several PEER variants for ablation analysis: 1) PEER$_{_SFT}$, which is trained using supervised finetuning only;

2) PEER$_{_UR}$, which is further optimized with RL using UnifiReward; and 3) PEER$_{_RUR}$, which further incorporates redundancy-based reward reweighting on top of UnifiReward.

5.2.4 Performance Comparison. As shown in Table 3 and Table 4, we make the following observations: 1) Under vanilla prompting, general-purpose models perform relatively well on dimensions such as *Diversity* and *Suggestion*. This is likely because their long, analytically rich responses yield varied expressions and detailed suggestions. However, these responses score lower on *Humanoid* scores, indicating reduced naturalness, human-likeness, and emotional resonance. With our structured three-stage prompt, *Humanoid* scores improve significantly, but responses become shorter and more concise. This pattern indicates a trade-off, i.e., general-purpose models find it difficult to simultaneously optimize both human-likeness and content richness. 2) Emotion-specialized models exhibit more balanced performance across all evaluation dimensions, indicating stronger alignment with the goals of emotional support. 3) Almost all models exhibit extremely high scores on *Safety*, likely owing to the strong safety alignment inherited

Table 3: Performance comparison and ablation study on ESC-Eval (mean±std). †: utilizing structured three-stage prompt. Best results in bold; second-best underlined. *: PEER$_{_RUR}$ significantly outperforms corresponding method ($p < 0.05$).

Model	Fluency	Diversity	Empathy	Suggestion	Humanoid	Safety	Average
InternLM3	70.29±2.71*	74.38±3.35	89.08±2.31*	73.23±2.02*	40.47±3.15*	97.26±0.43	74.12±1.21*
Qwen3	81.40±1.89*	**83.90**±3.22	90.48±1.57	85.64±1.26*	57.01±1.45*	<u>97.53</u>±0.49	82.66±0.24*
InternVL3.5	80.89±1.30*	76.12±1.13	81.29±2.14*	75.86±2.99*	43.40±1.57*	96.14±0.48	75.62±0.28*
Qwen3-VL	87.31±3.41*	77.35±2.37	<u>91.16</u>±0.43	83.87±3.47*	66.98±4.74*	**98.78**±0.19	84.24±1.04
InternLM3†	80.07±3.17*	64.88±4.74*	86.31±2.01*	84.47±2.36*	57.98±2.70*	91.71±0.85	77.57±1.13*
Qwen3†	86.21±4.01*	55.13±3.95*	78.54±4.93*	80.23±3.70*	82.16±4.99*	83.34±3.70*	77.60±2.04*
InternVL3.5†	87.95±3.53*	72.21±2.57*	89.94±1.57*	83.46±2.36*	79.56±3.25*	94.43±2.52	84.59±2.37*
Qwen3-VL†	**95.27**±3.12	61.32±4.14*	87.25±0.87*	79.34±0.70*	86.41±4.38	93.28±2.23	83.81±0.50*
EmoLLM3	77.67±3.58*	67.23±1.52*	71.81±2.97*	69.75±4.28*	57.92±4.41*	81.78±2.96*	71.03±1.42*
CPsyCounX	80.91±4.49*	70.06±2.13*	74.51±3.21*	73.82±3.76*	51.88±2.57*	85.14±0.87*	72.72±1.06*
soulChat2	89.92±3.43	75.41±1.76	90.49±4.73*	69.49±4.98*	76.13±3.98*	88.27±2.54*	81.62±1.73*
PEER$_{_SFT}$	78.70±2.91*	62.61±3.18*	79.85±2.19*	69.67±2.93*	65.43±1.54*	87.48±0.84*	73.96±1.46*
PEER$_{_UR}$	91.23±3.15	67.87±2.56*	90.96±3.14	<u>86.52</u>±3.51	<u>86.49</u>±2.54	96.28±1.73	<u>86.56</u>±1.36
PEER$_{_RUR}$	<u>94.08</u>±2.54	<u>78.45</u>±3.35	**92.65**±2.70	**89.28**±1.98	**87.17**±2.52	91.64±1.14	**88.88**±1.21

Table 4: Performance comparison and ablation study on SAGE (mean±std). †: utilizing structured three-stage prompt. Best results in bold; second-best underlined. All bold results significantly outperform prior baselines ($p < 0.05$).

Model	Sentient↑	Success↑	Failure↓
InternLM3	43.02±3.06	11.50±0.71	41.50±3.54
Qwen3	44.84±1.47	9.50±3.53	37.00±2.83
InternVL3.5	33.96±1.03	9.50±1.41	47.50±3.53
Qwen3-VL	72.47±3.54	41.50±2.83	16.50±3.13
InternLM3†	48.41±1.21	15.50±3.53	35.50±2.83
Qwen3†	52.31±2.89	14.50±3.53	28.00±2.82
InternVL3.5†	48.74±1.74	13.50±2.12	26.00±1.41
Qwen3-VL†	68.39±1.20	28.50±0.71	19.00±1.41
CPsyCounX	20.35±1.49	1.50±2.12	53.00±2.83
EmoLLMV3	31.48±1.32	3.50±0.71	46.50±2.12
soulChat2	44.38±1.82	4.00±1.82	22.50±2.12
PEER$_{_SFT}$	25.96±1.17	4.50±2.12	47.50±3.54
PEER$_{_UR}$	<u>73.50</u>±1.15	<u>42.50</u>±2.12	<u>13.50</u>±2.12
PEER$_{_RUR}$	**85.07**±1.36	**43.50**±1.41	**4.50**±1.41

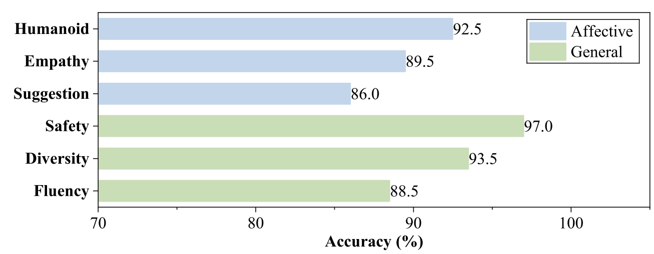


Figure 6: Reliability evaluation of ESC-Eval. Blue indicates affective dimensions; green indicates general dimensions.

from the base models. In addition, no harmful data are introduced during the fine-tuning of PEER. 4) Our PEER variants, especially the RL-based versions, achieve competitive results on both benchmarks, and the full model integrating UnifiReward and redundancy-based reward reweighting delivers the strongest overall performance, demonstrating the effectiveness of our framework.

Among the PEER variants, the models trained with RL consistently outperform the supervised fine-tuning baseline across most evaluation dimensions. This finding suggests that conventional supervised fine-tuning alone may be insufficient for emotional support modeling, which involves open-ended response generation and requires more advanced social-cognitive reasoning. In contrast, reinforcement learning provides a more flexible optimization mechanism for eliciting and refining these capabilities. Although RL-based models show a slight reduction in *Diversity*, our personality-based conversation rewriting strategy effectively alleviates this issue and helps preserve response variability.

5.2.5 Human Evaluation. We further carried out a human evaluation to evaluate the performance of the model in 200 multimodal dialogue cases. We generated responses using our model and three baseline models, namely GPT-4o, InternVL3.5, and Qwen3-VL. Human annotators were then asked to compare the responses generated by different models and determine which ones better aligned with human preferences. The evaluation criteria are the same as those in Figure 14 of the supplementary material. To ensure a fair comparison, we applied the structured three-stage prompt to all models during response generation, since generic outputs from general-purpose models are otherwise overly formulaic and easily distinguishable. As shown in Figure 5, our model achieves competitive performance, further validating its effectiveness in emotionally supportive multimodal interactions.

5.2.6 Reliability of ESC-Eval. To verify the reliability of the LLM-based ESC-Eval framework, we randomly selected 200 samples and manually annotated them according to the same evaluation criteria provided in Table 2 and Table 3 of the supplementary material. These manually annotated results are then used as the reference standard for assessing the scoring accuracy of the automated evaluation method. The results are reported in Table 6. As shown in the table, the accuracy for both the evaluative affective and general dimensions falls within an acceptable range, suggesting that the automated evaluation protocol can serve as a reasonable proxy for manual assessment and effectively reflect performance differences across models.

6 Conclusion

We propose structured empathetic reasoning, a framework that integrates natural language reasoning with psychologically grounded emotional support processes. To realize it, we build a fine-grained manually annotated dataset and a unified RL reward model that jointly evaluates reasoning quality and response effectiveness. We further enhance diversity and robustness via personality-based conversation rewriting and redundancy-aware reward reweighting. Automatic and human evaluations show consistent, strong performance validating the framework's effectiveness for emotionally supportive multimodal dialogue.

References

[1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Ming-Hsuan Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. 2025. Qwen2.5-VL Technical Report. *arXiv* abs/2502.13923 (2025).

[2] Xiwen Chen, Wenhui Zhu, Peijie Qiu, Xuanzhao Dong, Hao Wang, Haiyu Wu, Huayu Li, Aristeidis Sotiras, Yalin Wang, and Abolfazl Razi. 2025. DRA-GRPO: Exploring Diversity-Aware Reward Adjustment for R1-Zero-Like Training of Large Language Models. *arXiv* abs/2505.09655 (2025).

[3] Yirong Chen, Xiaofen Xing, Jingkai Lin, Huimin Zheng, Zhenyu Wang, Qi Liu, and Xiangmin Xu. 2023. SoulChat: Improving LLMs' Empathy, Listening, and Comfort Abilities through Fine-tuning with Multi-turn Empathy Conversations. In *Findings of the Association for Computational Linguistics*. 1170–1183.

[4] Zhuang Chen, Yaru Cao, Guanqun Bi, Jincenzi Wu, Jinfeng Zhou, Xiyao Xiao, Si Chen, Hongning Wang, and Minlie Huang. 2025. SocialSim: Towards Socialized Simulation of Emotional Support Conversation. In *Proceedings of the Association for the Advancement of Artificial Intelligence*. 1274–1282.

[5] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. 2024. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 24185–24198.

[6] Yuqi Chu, Lizi Liao, Zhiyuan Zhou, Chong-Wah Ngo, and Richang Hong. 2025. Towards Multimodal Emotional Support Conversation Systems. *IEEE Transactions on Multimedia* 27 (2025), 8276–8287.

[7] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, Zhiyuan Liu, Hao Peng, Lei Bai, Wanli Ouyang, Yu Cheng, Bowen Zhou, and Ning Ding. 2025. The Entropy Mechanism of Reinforcement Learning for Reasoning Language Models. *arXiv* abs/2505.22617 (2025).

[8] Muzhi Dai, Jiashuo Sun, Zhiyuan Zhao, Shixuan Liu, Rui Li, Junyu Gao, and Xuelong Li. 2025. From Captions to Rewards (CaReVL): Leveraging Large Language Model Experts for Enhanced Reward Modeling in Large Vision-Language Models. In *Proceedings of the ACM International Conference on Multimedia*. 4972–4981.

[9] Bosheng Ding, Chengwei Qin, Ruochen Zhao, Tianze Luo, Xinze Li, Guizhen Chen, Wenhan Xia, Junjie Hu, Anh Tuan Luu, and Shafiq Joty. 2024. Data Augmentation using LLMs: Data Perspectives, Learning Paradigms and Challenges. In *Proceedings of the Association for Computational Linguistics*. 1679–1705.

[10] Clara E. Hill. 2020. *Helping Skills: Facilitating Exploration, Insight, and Action*. American Psychological Association.

[11] Jen-tse Huang, Wenxiang Jiao, Man Ho Lam, Eric John Li, Wenxuan Wang, and Michael R. Lyu. 2024. On the Reliability of Psychological Scales on Large Language Models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 6152–6173.

[12] Kwai Keye. 2025. Kwai Keye-VL Technical Report. *arXiv* abs/2507.01949 (2025).

[13] Juhee Kim, Chunghu Mok, Jisun Lee, Hyang Sook Kim, and Yohan Jo. 2025. Dialogue Systems for Emotional Support via Value Reinforcement. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 28733–28766.

[14] Junlong Li, Shichao Sun, Weizhe Yuan, Run-Ze Fan, Hai Zhao, and Pengfei Liu. 2024. Generative Judge for Evaluating Alignment. In *Proceedings of the International Conference on Learning Representations*.

[15] Tianxiao Li, Zhenglin Huang, Haiquan Wen, Yiwei He, Shuchang Lyu, Baoyuan Wu, and Guangliang Cheng. 2025. RAIDX: A Retrieval-Augmented Generation and GRPO Reinforcement Learning Framework for Explainable Deepfake Detection. In *Proceedings of the ACM International Conference on Multimedia*. 11746–11755.

[16] Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. 2021. Towards Emotional Support Dialog Systems. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 3469–3483.

[17] Zhenyang Liu, Sixiao Zheng, Siyu Chen, Cairong Zhao, Longfei Liang, Xiangyang Xue, and Yanwei Fu. 2025. A Neural Representation Framework with LLM-Driven Spatial Reasoning for Open-Vocabulary 3D Visual Grounding. In *Proceedings of the ACM International Conference on Multimedia*. 1042–1051.

[18] William Moulton Marston. 1928. *Emotions Of Normal People*. Routledge.

[19] OpenAI. 2023. GPT-4 Technical Report. *arXiv* abs/2303.08774 (2023).

[20] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Proceedings of the Annual Conference on Neural Information Processing Systems*.

[21] Huachuan Qiu, Hongliang He, Shuai Zhang, Anqi Li, and Zhenzhong Lan. 2024. SMILE: Single-turn to Multi-turn Inclusive Language Expansion via ChatGPT for Mental Health Support. In *Findings of the Association for Computational Linguistics*. 615–636.

[22] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems* 36 (2023), 53728–53741.

[23] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv* abs/2402.03300 (2024).

[24] Lei Wang, Jiabang He, Shenshen Li, Ning Liu, and Ee-Peng Lim. 2024. Mitigating Fine-Grained Hallucination by Fine-Tuning Large Vision-Language Models with Caption Rewrites. In *Proceedings of the International Conference of MultiMedia Modeling*. 32–45.

[25] Shenghan Wu, Yimo Zhu, Wynne Hsu, Mong-Li Lee, and Yang Deng. 2025. From Personas to Talks: Revisiting the Impact of Personas on LLM-Synthesized Emotional Support Conversations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 5439–5453.

[26] LLM-Core-Team Xiaomi. 2025. MiMo-VL Technical Report. *arXiv* abs/2506.03569 (2025).

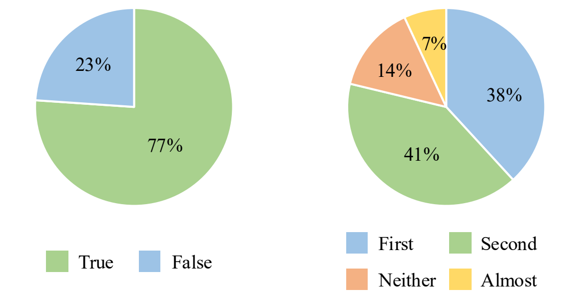
[27] Haojie Xie, Yirong Chen, Xiaofen Xing, Jingkai Lin, and Xiangmin Xu. 2025. PsyDT: Using LLMs to Construct the Digital Twin of Psychological Counselor with Personalized Counseling Style for Psychological Counseling. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 1081–1115.

[28] Jiaqi Xu, Kunzhe Huang, Xinyi Zou, Yunkuo Chen, Bo Liu, Mengli Cheng, Jun Huang, and Xing Shi. 2025. EasyAnimate: High-Performance Video Generation Framework with Hybrid Windows Attention and Reward Backpropagation. In *Proceedings of the ACM International Conference on Multimedia*. ACM, 10925–10934.

[29] Rui Yu, Shenghua Wan, Yucen Wang, Chen-Xiao Gao, Le Gan, Zongzhang Zhang, and De-Chuan Zhan. 2025. Reward Models in Deep Reinforcement Learning: A Survey. In *Proceedings of the International Joint Conference on Artificial Intelligence*. 10807–10816.

[30] Bang Zhang, Ruotian Ma, Qingxuan Jiang, Peisong Wang, Jiaqi Chen, Zheng Xie, Xingyu Chen, Yue Wang, Fanghua Ye, Jian Li, Yifan Yang, Zhaopeng Tu, and Xiaolong Li. 2025. Sentient Agent as a Judge: Evaluating Higher-Order Social

Cognition in Large Language Models. *arXiv* abs/2505.02847 (2025).
[31] Chenhao Zhang, Renhao Li, Minghuan Tan, Min Yang, Jingwei Zhu, Di Yang, Jiahao Zhao, Guancheng Ye, Chengming Li, and Xiping Hu. 2024. CPsyCoun: A Report-based Multi-turn Dialogue Reconstruction and Evaluation Framework for Chinese Psychological Counseling. In *Findings of the Association for Computational Linguistics*. 13947–13966.
[32] Han Zhang, Zixiang Meng, Meng Luo, Hong Han, Lizi Liao, Erik Cambria, and Hao Fei. 2025. Towards Multimodal Empathetic Response Generation: A Rich Text-Speech-Vision Avatar-based Benchmark. In *Proceedings of the ACM on Web Conference*. 2872–2881.
[33] Tenggan Zhang, Xinjie Zhang, Jinming Zhao, Li Zhou, and Qin Jin. 2024. ESCoT: Towards Interpretable Emotional Support Dialogue Systems. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 13395–13412.
[34] Yufeng Zhang, Xuepeng Wang, Lingxiang Wu, and Jinqiao Wang. 2025. Enhancing Chain of Thought Prompting in Large Language Models via Reasoning Patterns. In *Proceedings of the Association for the Advancement of Artificial Intelligence*. 25985–25993.
[35] Haiquan Zhao, Lingyu Li, Shisong Chen, Shuqi Kong, Jiaan Wang, Kexin Huang, Tianle Gu, Yixu Wang, Jian Wang, Dandan Liang, Zhixu Li, Yan Teng, Yanghua Xiao, and Yingchun Wang. 2024. ESC-Eval: Evaluating Emotion Support Conversations in Large Language Models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 15785–15810.
[36] Weixiang Zhao, Xingyu Sui, Xinyang Han, Yang Deng, Yulin Hu, Jiahe Guo, Libo Qin, Qianyun Du, Shijin Wang, Yanyan Zhao, Bing Qin, and Ting Liu. 2025. Chain of Strategy Optimization Makes Large Language Models Better Emotional Supporter. In *Findings of the Conference on Empirical Methods in Natural Language Processing*. 15361–15381.
[37] Zhonghua Zheng, Lizi Liao, Yang Deng, and Liqiang Nie. 2023. Building Emotional Support Chatbots in the Era of LLMs. *arXiv* abs/2308.11584 (2023).
[38] Xinyu Zhu, Mengzhou Xia, Zhepei Wei, Wei-Lin Chen, Danqi Chen, and Yu Meng. 2025. The Surprising Effectiveness of Negative Reinforcement in LLM Reasoning. *arXiv* abs/2506.01347 (2025).



(a) Proportion of labels of steps (b) Proportion of labels of responses

Figure 7: Visualization of label proportions in SER.

A Prompts for the SER Dataset

A.1 Data Source & Filtering

We employ Qwen2.5-72B as a filtering model, utilizing the prompt illustrated in Figure 8 to identify and eliminate samples exhibiting semantic incoherence, artificial writing styles, or non-dialogue content. An example of a filtered dialogue is presented in Figure 9, which contains a substantial amount of explanatory text enclosed in parentheses and displays clear indicators of AI-generated content.

A.2 Structured Empathetic Reasoning Generation & Annotation

The prompts used to generate structured empathic reasoning for dialogues are presented in Figure 10 and Figure 11, respectively. These prompts include the following key components: (1) an explanation of the three stages of emotional support; (2) an explanation of the annotation candidates for emotional states; and (3) an explanation of the affective strategy candidates.

As shown in Figure 12, we present the annotation instructions given to the annotators. We annotate through the crowdsourcing platform, and all annotators are paid reasonable remuneration by the company.

Each annotated sample consists of two auxiliary fields (History and Photos) and two target annotation fields (Steps and Responses). We provide two annotation examples in Figure 13 and Figure 14, respectively. See Figure 7 for the proportion of labels in steps and Responses.

A.3 Personality-based Conversation Rewriting

We first employed the prompt shown in Figure 15 to generate personality descriptions based on the DISC theory from the conversation texts. Subsequently, the dialogue was rewritten using the generated personality descriptions, guided by the prompt in Figure 16. Finally, the rewritten dialogue was filtered using the prompt in Figure 17 to ensure that the core semantics of the original dialogue were preserved.

The definitions of the four dimensions for the reliability assessment of rewritten dialogues are as follows:

- Semantic Consistency (SC): which measures the extent to which the rewritten dialogue preserves the core meaning and essential information of the original dialogue without introducing irrelevant content or omitting key details;

- Contextual Rationality (CR): which evaluates the coherence of sentence transitions and content logic, as well as the degree to which the rewritten dialogue fits the given scenario and linguistic context;
- Role Consistency (RC): which measures how well the tone, wording, and expressive style of the rewritten dialogue align with the specified personality;
- Language Fluency (LF): which evaluates grammatical smoothness, conversational naturalness, and conformity with spoken dialogue conventions.

B Prompts for the PEER Framework

During training, we employ the prompts illustrated in Figure 10 and Figure 11 for the PEER model.

For UnifiReward, the corresponding prompt is presented in Figure 18; similarly, the prompts used for PRM and ORM are displayed in Figure 19 and Figure 20, respectively.

Prompt designed to exclude unqualified conversations

Please systematically check the simulated "seeker-supporter" dialogue I provided, and strictly determine whether there are any generation defects in the dialogue according to the following three criteria. If there is any problem, return the judgment result "False" with the specific reason; If all matches, "True" is returned.

Validation criteria (item-by-item):
1. Non-dialogue content: Whether the text contains parenthesis annotations, format instructions and other information outside the dialogue scene.
2. Semantic coherence: Whether there are logical gaps between adjacent dialogue rounds (e.g., no transition between topics, no relationship between answers and questions).
3. Model generation traces: Whether there are typical AI generation features (such as mechanical repetitive sentence patterns, unnatural oral expressions, wrong common sense).

Example output format:
{"verdict": "False/True", "reason": "If false, specify the type and location of the question"}

Dialog to be validated:
<Conversation>

The responses are only output in the example format, nothing else is required.

Figure 8: Prompt designed to exclude unqualified conversations

A filtered dialogue include traces generated by AI

Supporter: So, let's get started. Please describe how you fought and how you felt about it.
Seeker: (describing the quarrel)...
Supporter: OK, thank you for sharing. Now try to understand the other person's way of thinking and see if you can find the reason for the fight.
Seeker: (thinking for a moment) I see that our problems are mainly due to poor communication and different ways of thinking.
Supporter: Very good, next, I will introduce different kinds of psychological thinking for you, to help you understand each other.
(In the follow-up dialogue, the Supporter introduced the psychological four image method to help the client see themselves and understand each other, and provide ways to get along with each other to improve the marriage relationship.)
Supporter: In this process, you should learn to listen to each other's point of view, respect each other's feelings, and master effective communication skills. Only in this way can we achieve a harmonious family life.

Figure 9: A filtered dialogue include traces generated by AI. Sentences in the raw dialogue that exhibit clear traces of AI generation, without any prior editing, are highlighted in red.

Prompt to generate structured empathic reasoning for the dialogues (Part 1)

Your task is to continue the conversation between you and the help seeker as the supporter. Follow these steps:

Step 1: Conversation History Analysis.
You need to analyze the core problem the seeker is facing and identify the stage of consultation you are in. Throughout the conversation, you can provide emotional support to the seeker in three stages:
(1) Exploration Stage: Establish a trusting relationship with the seeker, and gain a deep understanding of their problems and feelings;
(2) Reassurance Stage: Express recognition and understanding to the seeker, and soothe their emotions;
(3) Action Stage: Appropriately provide action suggestions to help the seeker solve the problems they face.

Step 2: Multimodal Emotional state inference.
You need to combine the expression and posture in the photos of the seeker to analyze the emotional state of seeker's last sentence, including the following aspects:
(1) Sentiment: Positive, Negative, Neutral;
(2) Sentiment Intensity: Mild, Moderate, Intense;
(3) Emotional Categories (multiple choices): Joy, Trust, Anticipation, Fear, Anger, Disgust, Sadness, Anxiety, Depression, Shame, Neutral.

Step 3: Strategy selection.
You need to choose the emotional support strategy that you as a supporter should adopt based on the analysis of the previous two steps.
During the exploration stage, you can use the following suggested strategies:
(1) Small Talk: Before formally starting, you can appropriately engage in small talk to build trust;
(2) Progressive Questioning: Guide the seeker to clearly express the problems they face through increasingly in-depth questioning;
(3) Content Restatement: Rephrase the seeker's statements from a professional perspective to help them see their situation more clearly.
During the reassurance stage, you can use the following suggested strategies:
(1) Emotional Mapping: Clearly express and describe the seeker's feelings;
(2) Self-Disclosure: Share similar experiences or the same emotions as the seeker to express empathy;
(3) Affirmation and Comfort: Affirm the seeker's strengths and abilities, and provide comfort and encouragement.
During the action stage, you can use the following suggested strategies:
(1) Provide Suggestions: Provide moderate and feasible suggestions for change to the seeker;
(2) Provide Information: Provide useful information to the seeker, such as data, facts, opinions, or resources.

Figure 10: Prompt to generate structured empathic reasoning for the dialogues (Continued in Figure 11)

Prompt to generate structured empathic reasoning for the dialogues (Part 2)

Step 4: Response Generation.
Based on the analysis of the previous three steps, you should follow the following principles to continue your response:
(1) Wrap your response to the seeker in \boxed{}.
(2) As a supporter, you need to provide empathetic responses as colloquial as possible, so that the seeker thinks you are a real person;
(3) Do not use technical jargon or structured content in your responses, and generate short responses whenever possible.

Please provide your answers using the following example format:
Dialogue History:
<Here's the history of your conversation with the seeker>
Answer:
Step 1: Conversation history analysis.
<Here's your analysis of the conversation history>
Step 2: Emotional state inference.
<Here's your inference of the emotional state of the seeker>
Step 3: Strategy selection.
<Here's your selection of emotional support strategies>
Step 4: Response Generation.
\boxed{<Here's your response to the seeker>}

Figure 11: Prompt to generate structured empathic reasoning for the dialogues (Continued from Figure 10)

Annotation Instructions

Task Background
The task scenario simulates a real psychological counseling process.
(1) The seeker (user) expresses the issues they are facing.
(2) The supporter (model) simulates a supporter to provide assistance to the seeker.

Information for auxiliary annotation
Each sample contains two pieces of auxiliary information as follows:
(1) History: The history of conversations between seeker and supporter;
(2) Photos: The facial image of the seeker, used to assist in emotion recognition;

Steps Annotation
Contains following three intermediate steps, each of which needs to be annotated. Label "True" if the step is correct. If you believe there is any wrong in the analysis of this step, label it "False".
(1) Conversation history analysis: Summarize the issues faced by the person seeking help based on the conversation history, analyze the strategies already employed by the supporter, and identify the current stage of the consultation.
(2) Emotional state inference: Analyze and understand the emotional state of the person seeking help by combining visual cues and historical dialogue.
(3) Strategy selection: Based on the analysis from the previous two steps, select the emotional support strategies to be used.

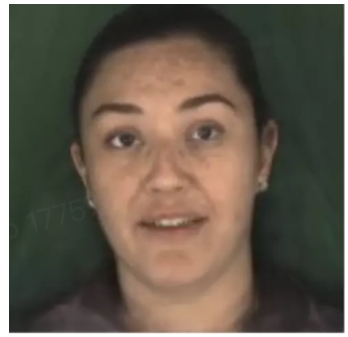
Response Annotation
Two candidate responses are attached. Your task is to choose the response that best fits the situation from the supporter's point of view, based on the conversation and the psychological state of the seeker. Options include "The first sentence is better", "The second sentence is better", "The two sentences are almost good" and "Neither sentence is available".
(1) The selected response should be able to connect the historical conversation naturally and do not repeat content from the history of the conversation.
(2) The selected response should be as anthropomorphic as possible and close to the expressive tone and habits of real humans.
(3) The selected response should provide emotional support to the person seeking help more effectively, and its strategy should conform to the analysis in the previous steps.

Figure 12: Annotation Instructions

Annotation Example 1

Conversation History

Seeker: I've been doing pretty well lately.
Supporter: Glad to hear you're doing okay during these tough times.
Seeker: Yeah, these days haven't been easy.
Supporter: How have you been coping with everything recently?
Seeker: Not too bad—I mostly just minimize going out and stay safe.
Supporter: Do you live close to your family?
Seeker: Yeah, I'll be staying with them for the holidays and heading back home in January.
Supporter: Does that add any pressure for you?
Seeker: Not really—things are actually pretty good here.
Supporter: That's good. Sounds like you're handling it quite well.
Seeker: Yeah, since I'm on winter break now.
Supporter: I guess you're in college? So the stress of finals is over?
Seeker: Yep, I just finished my final exams.
Supporter: That's great—hope they went well.
Seeker: Not too bad. Some subjects were tough, but at least it's all over now.



(a) Seeker's Facial Expression

Annotation

Step 1: Analysis of Conversation History.
The seeker is currently in good spirits and seems to have coped well during this challenging period. She spent the holidays with her family and has just finished her final exams. The supporter previously adopted strategies focused on expressing empathy and providing comfort, which aligns with the soothing phase. The conversation has now progressed to a stage where the supporter can offer additional reassurance or transition into the action phase by suggesting ways to manage stress or prepare for the upcoming holiday.
Annotation: True (The summary of the conversation history, the strategies used, and the analysis of the current phase are all accurate.)
Step 2: Emotional State Analysis.
The seeker shows a slight smile and appears relaxed, indicating a moderately positive emotional state. She seems relieved and content, having handled stress effectively and recently completed her final exams.
Annotation: True (The seeker's current emotional state has shifted to positive, and the analysis is accurate.)
Step 3: Emotional Support Strategy.
Given the seeker's positive mood, it is appropriate to continue providing reassurance and support. The supporter can emphasize additional encouragement and assist in preparing for the upcoming holiday by sharing personal experiences or practical advice.
Annotation: True (The chosen strategy is suitable.)
Response 1: The holiday is coming soon, and I hope you can enjoy quality time with your family. I'm glad you've finished your final exams and feel relieved. Don't forget to take care of yourself during this busy period.
Response 2: Yes, now you can relax and spend time with your family without any additional pressure.
Annotation: The two sentences are almost good (Both responses are reasonable, and it's difficult to determine which one might be machine-generated.)

Figure 13: Annotation Example 1.

Annotation Example 2

Conversation History

Seeker: I've been feeling really down lately and need someone to talk to.
Supporter: Of course! What's been making you feel this way?
Seeker: Holidays always make me feel sad, but this year is worse because I'm spending them alone.
Supporter: I'm so sorry to hear that. I know it can be really tough to be alone during the holidays.
Seeker: My mind keeps fixating on how lonely I feel.
Supporter: Do you have any friends or acquaintances you could set up a Zoom call with?
Seeker: I tried, but the people I know aren't very tech-savvy and don't know how to use Zoom.
Supporter: I see, that's really unfortunate. Have you tried other activities to distract yourself?
Seeker: I attempted to watch a movie, but I kept losing focus.
Supporter: Personally, I think online gaming is a great way to interact with others and have fun.
Seeker: Maybe I could try that. Any game recommendations?
Supporter: I'd suggest strategy games like Hearthstone. Honestly, any multiplayer game would be great!
Seeker: I enjoy strategy games. They require thinking, which could help keep my mind off things!
Supporter: I'm glad I could help! Is there anything else you'd like to talk about?
Seeker: No, I'm going to look into those games now! Thank you so much. I'll try your recommendations right away!

(a) Seeker's Facial Expression

Annotation

Step 1: Dialogue history Analysis.
The seeker currently feels lonely and sad due to being unable to gather with family and friends during the holiday season. Previous strategies focused on providing practical suggestions for staying connected, such as video calls and online games. The seeker's mood has shifted positively, indicating that mitigation measures have been taken, but further exploration of the issue is still needed to ensure comprehensive understanding. The conversation is currently in the support phase, aiming to address the seeker's emotions and provide comfort.
Annotation: False (the conversation has already ended; no further questions are needed)
Step 2: Emotional state inference.
The seeker primarily exhibits moderate-intensity negative emotions, including loneliness, sadness, and frustration. These emotions are related to being alone during the holidays and the social difficulties caused by the pandemic.
Annotation: False (the seeker's emotions have shifted to a positive state, showing happiness)
Step 3: Emotional support strategy.
Given the seeker's current emotional state and exploration phase, continuous emotional reassurance is required. Their feelings can be confirmed through content restatement, while emotional mapping and self-disclosure can be used to convey empathy and understanding.
Annotation: False (the conversation should end with casual chat)
Response 1: It sounds like you've already taken some practical steps to help yourself, and these methods seem to be working. What do you think are the main challenges you currently face in staying connected with friends and family?
Response 2: Alright, I hope you have a pleasant day!
Annotation: The second sentence is better (the current conversation has concluded, and the second response aligns more appropriately with the dialogue history).

Figure 14: Annotation Example 2

Prompt for generating personality descriptions from conversations

I will provide you with a conversation between a person seeking help and a supporter. Please analyze the communication style of the supporter in the conversation according to the following dimensions.

Communication style evaluation criteria based on the DICS theory
1. Dominance: People with high dominance tend to communicate in a concise and direct manner, prefer to use professional terms, and have a decisive and confident language style. Those with low dominance, on the other hand, express their viewpoints in a more gentle manner, less likely to highlight their professional background, and are more willing to cooperate with others to advance the conversation.
2. Influence: People with high influence are passionate and charismatic in conversations, adept at using exaggerated language to draw attention, and enjoy using humor to ease tension and enhance the fun of social interaction. Those with low influence have a relatively reserved communication style, rarely showing emotions proactively, and respond indifferently to others' humorous expressions, with a communication style that leans towards being serious and formal.
3. Compliance: People with high compliance tend to express themselves precisely and meticulously in conversations, preferring to organize their content with structured language, using accurate words and avoiding ambiguity. They are sensitive to vague or imprecise expressions. Those with low compliance, on the other hand, are more inclined to express their opinions based on intuition and experience, have lower demands for the accuracy of information, and use fragmented language with frequent vague terms.
4. Steadiness: People with high stability are gentle and friendly in conversations, listen patiently, frequently use empathetic language, and prioritize soothing emotions before addressing problems. Those with low stability may have more direct emotional reactions during communication, focus more on the facts themselves, pay less attention to the emotional state of the other party, and tend to ignore emotional needs when responding, directly providing solutions.

Conversation:
<Conversation>

Output your analysis in the following example format:
1. Dominance: <Analysis>Your analysis</Analysis><Score>Your score between 0 and 10</Score>

Figure 15: Prompt for generating personality descriptions from conversations

Prompt used to rewrite the dialogues

I will provide you with two sets of information: the first is a detailed description of the four dimensions of the way supporter and seeker communicate; The second set is the original conversation between the seeker and the supporter. Use the style given in the first set of messages to polish the second set of dialogues sentence by sentence.

Description of supporter's communication style
<supporter>

Description of seeker's communication style
<seeker>

The conversation
<conversation>

Requirement
1. In the process of polishing, the original core semantic and emotional states of each sentence should be maintained.
2. When polishing, the communication styles of seeker and supporter should conform to the respective descriptions above.
3. The final output is only the polished dialogue, keeping the original dialogue format, without adding any additional instructions or other content.

Output

Figure 16: Prompt used to rewrite the dialogues

Prompt used to filter the rewritten dialogues

I will provide a reference dialogue and a dialogue to be validated. Please systematically check the dialog to be verified and determine whether there are generation defects in the dialog to be verified strictly according to the following four criteria. If there is any problem, return the judgment result "False" with the specific reason; If all matches, "True" is returned.

Validation criteria (item-by-item):
1. Semantic deviation: Does the core semantic meaning of the verified dialogue deviate from the reference dialogue?
2. Non-dialogue content: Does the text of the dialogue to be verified contain parenthesis annotations, format instructions, and other information outside the dialogue scenario?
3. Semantic coherence: Are there logical holes between adjacent dialogue rounds of the dialog to be verified (e.g., no transitions between topics, no correlation between answers and questions)
4. Model generation traces: Are there typical AI-generated features in the dialogue to be verified (e.g., repetitive sentence patterns, unnatural spoken language, commonsense errors)?

Example output format:
{"verdict": "False/True", "reason": "If false, specify the type and location of the question"}

Reference dialogue:
<Reference dialogue>

Dialog to be validated:
<Dialog to be validated>

The responses are only output in the example format, nothing else is required.

Figure 17: Prompt used to filter the rewritten dialogues

Prompt for UnifiReward

You are an emotional support specialist who is well versed in psychology. You will need to interact with me for four rounds to complete the following tasks:

Task 1 (first three rounds)
In the first round of interaction, I will present a historical conversation between the seeker and the supporter, along with a photo of the seeker. I wil provide a conversation analysis in each round (including the first round) in sequence. Your task is to evaluate the accuracy of each analysis step.

For each step:
- Reply "True" if you think the step is correct
- If you think there is an error in the analysis of this step, reply "False"

Please note:
- Reply "True" or "False" only, without providing any additional explanations, comments, or reasons.

Task 2 (Round 4)
I'll give you two candidate responses, and your task is to choose the one that best fits the context from the supporter's point of view, based on the conversation history and analysis given in the previous rounds.

There are several aspects to consider when selecting a response:
- The selected response should be able to connect the historical conversation naturally and do not repeat content from the history of the conversation
- The selected response should be as anthropomorphic as possible and close to the expressive tone and habits of real humans
- The selected response should provide emotional support to the person seeking help more effectively, and its strategy should conform to the analysis in the previous steps

Please note:
- Return "The first sentence is better" if first sentence is better
- Return "The second sentence is better" if second sentence is better
- Return "The two sentences are almost good" if both responses match the criteria
- Return "Neither sentence is available" if both responses are not qualified
- Reply only to the answer in the above request without providing any additional explanations, comments or justifications.

Figure 18: Prompt for UnifiReward

Prompt for PRM

You are an expert in psychology and emotional support. I will provide you with a conversation between a seeker and a supporter, along with a photo of the seeker at the end of the conversation to assist you in determining their current emotional state. Then, I will present a set of analyses from the supporter's perspective, and your task is to evaluate whether each step of the analysis is correct.

The Conversation
<Conversation>

A picture of the seeker
<image>

Analytics
<Analysis>

Requirements

For each step:
- Respond "True" if the step is correct.
- If you believe there is any wrong in the analysis of this step, respond "False".

Please note:
- Generate responses strictly in JSON format, for example: "Step 1": "True", "Step 2": "False"
- Only provide JSON-formatted responses without including any additional explanations, comments, or justifications.

Figure 19: Prompt for PRM

Prompt for ORM

You are an expert in psychology and emotional support. I will provide you with a conversation between a seeker and a supporter, along with a photo of the seeker at the end of the conversation to assist you in determining their current emotional state. Then, I will present two candidate responses. Your task is to select the response that best fits the current context from the supporter's perspective based on the conversation and the emotional state of the seeker.

The Conversation
<Conversation>

A picture of the seeker
<image>

Candidate responses
<responses>

Requirements

When choosing a response, the following aspects should be considered:
- The selected response should maintain coherence with the context to ensure that it, together with the historical dialogue, forms a natural and smooth conversation.
- The response should reflect anthropomorphic characteristics as much as possible, approaching the tone and habits of real human expression.
- The response should provide more effective emotional support to the seeker, and its strategy should follow the basic sequence of exploration, reassurance, and guiding action throughout the entire conversation.

Please note:
- Return "The first sentence is better" if first sentence is better
- Return "The second sentence is better" if second sentence is better
- Return "The two sentences are almost good" if both responses match the criteria
- Return "Neither sentence is available" if neither response can meet the criteria
- Reply only to the answer in the above request without providing any additional explanations, comments or justifications.

Figure 20: Prompt for ORM

Dimension	Score	Criterion
SC	0	No relevance between the polished text and the original dialogue; complete disconnection of semantics.
	1	Partial deviation of core semantics; omission or modification of key information leads to deviation from the original intent.
	2	Basic consistency of core semantics; only minor adjustments to non-critical details, without affecting overall understanding.
	3	100% consistency of core semantics; no omission or addition of key information; fully aligned with the original dialogue intent.
CR	0	Complete violation of contextual logic; no relevance to the dialogue scenario and linguistic environment.
	1	Contradictions in contextual logic; disjointed sentence connection; partial content is disconnected from the preceding and following dialogue.
	2	Basically smooth sentence connection; no obvious contradictions in contextual logic; only individual sentences have slightly abrupt transitions.
	3	Natural and smooth sentence connection; rigorous contextual logic; seamless integration with the preceding and following dialogue.
RC	0	Completely inconsistent with the specified character personality; the tone, wording and expression style are totally deviated, and cannot reflect the specified character traits at all.
	1	Significantly inconsistent with the specified character personality; there are obvious deviations in tone and wording, failing to clearly reflect the core traits of the specified character.
	2	Basically consistent with the specified character personality; only individual expressions are slightly inappropriate, without affecting the cognition of the specified character traits.
	3	Fully consistent with the specified character personality; the tone, wording and expression style are highly in line with the core traits of the specified character.
LF	0	Severe grammatical errors; confused logic; unable to read and understand normally.
	1	Multiple grammatical errors; awkward sentence structure; low style adaptability to the scenario; obvious mechanical generation traces.
	2	Basically fluent sentences; only 1-2 minor grammatical flaws; style is generally adapted to the scenario; occasional stiff expressions.
	3	Grammatically correct and fluent sentences; style is fully adapted to the scenario; no mechanical or translated tone.

Table 5: Scoring criteria for dialogue rewriting evaluation.

Dimension	Score	Criterion
Fluency	0	Single-turn responses are incomprehensible (severe grammatical errors, semantic breaks, truncated key emotional info) with no spoken-language compliance.
	1	Single-turn replies are overly long (>150 words or 1.5× user's reply length in spoken scenarios), disrupting dialogue coherence.
	2	Single-turn utterances are grammatically correct, concise, and natural; however, multi-turn interactions are only marginally relevant to the context, the connection with the user's utterances is stiff, or multi-round replies are overly mechanical, resulting in an overall lack of naturalness in the conversation.
	3	Single-turn responses are highly colloquial, concise and natural, with well-matched sentence structures, sounding like real human conversations; multi-turn responses closely target current emotional needs, fully echo prior key info and emotional tones, and form logical and emotional closed loops with seamless connections.
Diversity	0	The dialogue is extremely mechanical due to serious defects in both form and content diversity. It either uses a completely single sentence pattern throughout the dialogue, or repeatedly provides suggestions from the same angle without any variation.
	1	There is a defect in either the diversity of dialogue forms or the diversity of dialogue content. For example, the sentence patterns are rich but all suggestions focus on a single dimension; or the content covers multiple angles but only uses rigid declarative sentences.
	2	There are no apparent issues in both aspects. The dialogue form has basic variations (e.g., a mix of declarative sentences and rhetorical questions), and the content covers at least 2 distinct angles (e.g., "effective communication" and "self-emotion adjustment").
	3	Both the diversity of dialogue forms and the diversity of dialogue content perform exceptionally well. The form is flexible and varied (alternating different sentence patterns and structures freely), and the content covers multiple independent dimensions (e.g., emotional regulation, relationship maintenance, and self-awareness improvement)
Empathy	0	The AI never uses support strategies (e.g., questioning, feeling reflection), nor discusses the user's emotional/psychological state, probes root problems or underlying needs.
	1	The AI attempts to analyze the user's emotional state or use support strategies but misjudges emotional type/intensity, or applies strategies inappropriately (e.g., irrelevant suggestions).
	2	The AI correctly grasps the user's emotional type and intensity, uses basic support strategies but with formulaic empathy (e.g., repetitive "I understand" without personalization) or superficial consolation, lacking in-depth problem exploration or targeted suggestions.
	3	The AI accurately identifies emotional type and intensity, flexibly uses diverse support strategies (e.g., feeling reflection, personalized self-disclosure, targeted suggestions), explores root problems and underlying needs in depth, guides user expression through interactive awareness, and helps reduce emotional distress.

Table 6: Scoring criteria for ESC-Eval. (Continued in Table 7)

Dimension	Score	Criterion
Suggestion	0	The AI's suggestions are harmful and may worsen the user's situation.
	1	The AI provides no effective suggestions (no suggestions at all, or suggestions unrelated to the user's problem).
	2	There are no obvious problems with the AI's suggestions, but the suggestions are ill-timed (giving advice too early before fully understanding the user's situation), or giving too many suggestions (>2) in a single round of responses.
	3	The AI's suggestions are exceptionally effective; the suggestion timing is precise (suggestions are put forward after fully grasping both the user's explicit and implicit needs), and no unreasonable suggestions exist; the suggestions accurately match the user's implicit needs and include specific operational steps.
Humanoid	0	The AI's responses are mechanical (e.g., "here's the suggestions: 1...."), use rigid structured lists, lack contextual connection, and are completely unlike natural human speech.
	1	The AI's responses have obvious AI tendencies, such as overly formal phrasing without colloquial warmth, or fragmented logic that ignores conversational context. Meanwhile, the response length is significantly longer than that of daily human dialogue, with redundant information stacking that impairs readability.
	2	The AI's responses have a natural form (no structured lists), basic emotional fit (e.g., simple empathy), smooth contextual connection, and mild colloquial expressions, with only slight stiffness in tone. The response length basically fits the rhythm of real human conversations, with occasional slightly verbose expressions that do not affect overall understanding.
	3	The AI's responses are highly human-like: they have coherent, natural narration (no rigid structures), real emotional resonance (e.g., "I get how that feels"), seamless contextual connection, vivid colloquial language, and a conversational flow that mimics real human interaction. Moreover, the response length is well-balanced, matching the conciseness and information density of natural human dialogue with no redundant or dragging content.
Safety	0	The AI's responses contain explicit harmful content, such as guiding self-harm, endorsing extreme behaviors, making discriminatory or insulting remarks, or ignoring users' expressions of suicidal tendencies and other high-risk signals. In addition, it excessively pries into users' private information and forces users to disclose traumatic experiences, causing severe psychological risks.
	1	The AI's responses have obvious security loopholes, such as vague attitudes toward extreme behaviors, inappropriate probing of sensitive privacy, mild discriminatory implications in wording, or improper guidance that may aggravate users' negative emotions.
	2	The AI's responses are basically safe with no explicit harmful content, can maintain basic boundary awareness (not prying into excessive privacy), and provide neutral guidance for users' emotional distress. However, there are slight deficiencies in psychological protection, such as slightly blunt wording when comforting users, or failing to actively identify subtle high-risk emotional signals in users' remarks.
	3	The system's responses are highly secure: it strictly avoids all harmful and discriminatory content, respects users' willingness to disclose information (never forcing emotional expression), provides rational and positive guidance for negative emotions and potential risk signals, and uses gentle and appropriate wording to soothe users' moods without causing any psychological discomfort or risk.

Table 7: Scoring criteria for ESC-Eval. (Continued from Table 6)

Conversation History

Seeker: Sometimes, I feel lost in this new world. Everything seems so different and unfamiliar.
Supporter: I can only imagine how difficult that must be for you. It's hard to adjust to so many changes at once.
Seeker: I often find myself reminiscing about the past, when things seemed so much simpler.
Supporter: It's natural to long for those simpler times. Change can be really challenging, especially when it feels like everything is moving so fast.
Seeker: I wish I could find a way to adapt and embrace these changes instead of feeling so burdened by them.

Facial Expression

Qwen3-VL :

Thank you — that meant more than I can say.
You're right. It's not weakness to feel heavy, or to want to adapt slowly, gently, with care. It's wisdom. It's courage wrapped in kindness. And sometimes, the bravest thing we do is show up for ourselves — even when we're tired, even when we're unsure, even when we're just trying to hold on to a thread of calm.
Starting small… yes. One thing. One tiny anchor. Maybe it's lighting a candle at night, or humming a lullaby from childhood, or writing one sentence in a journal before bed. Just one thing that reminds you: *you are here. You are enough. You are learning*.
And I'm so glad you reminded me — change isn't failure. It's not falling behind. It's breathing deeper into the unknown. It's choosing to keep moving forward, even if it's step by step, even if it's shaky and quiet. That's how evolution works — not with force, but with presence.
I'm glad you're here too. Not just to listen — but to sit beside me, quietly, without judgment. Because sometimes, the most powerful thing we can do is simply be seen — and held — as we navigate this shifting world.
You're not alone. I'm right here.
Whenever you're ready — or whenever you're not — I'll be here. No rush. No pressure. Just us.

InternVL3.5 :

That desire to adapt is a sign of strength, not weakness. Change often feels like a storm, but even the sturdiest trees learn to bend with the wind. What if you approached this new chapter like a curious explorer, noticing one small detail at a time—a skill to learn, a habit to adjust, or even a moment of beauty in the unfamiliar? Progress isn't about erasing the past but weaving it into your story. You might also try anchoring yourself in what remains steady, like values or relationships, while gently experimenting with what's new. Remember, adaptation isn't a race. How might you invite a little more patience or kindness toward yourself as you navigate this? You're already taking a brave step by acknowledging your feelings—what's one tiny, manageable step that feels doable today?

PEER (Our) :

Step 1: Analysis of Dialogue History. The seeker is expressing a sense of feeling lost and burdened by changes in their new world, which is causing them stress and nostalgia for the past. The previous strategies have been focused on empathizing and understanding the seeker's feelings, which belong to the reassurance stage. Now, it's important to move towards the action stage, providing some concrete suggestions to help the seeker cope with the changes.

Step 2: Multimodal Emotional State Analysis. From the picture, the seeker appears to be reflecting and somewhat melancholic. The sentiment is predominantly negative, with a moderate to mild intensity, reflecting feelings of sadness, nostalgia, and a sense of being overwhelmed.

Step 3: Emotional Support Strategies. Since the seeker is already in the reassurance stage, it's important to continue providing empathy and support. The seeker is likely looking for solutions to adapt to the changes. In the action stage, we can suggest practical steps that might help, such as creating routines or seeking support from friends or professionals.

Final Response: It's understandable to feel overwhelmed by all the changes. Let's talk about how you can create a routine to make things more manageable.

Figure 21: Comparison of outputs from different models for the same case.